

# Legendre’s Conjecture in Function Fields: Full Monodromy, Twisted Root Varieties, and Explicit Bounds

Revised version (v6) of *The Geometry of Prime Vacuums*

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## Abstract

For a monic polynomial  $f \in \mathbb{F}_q[t]$  of degree  $d \geq 1$ , the *Legendre interval*  $\mathcal{I}_f = \{f^2 + s : \deg s \leq d\}$  is the natural function field analogue of  $[n^2, (n+1)^2]$ . That  $\mathcal{I}_f$  contains  $q^{d+1}/(2d) + O_d(q^{d+1/2})$  irreducible polynomials is a special case ( $\epsilon = 1/2$ ) of the short-interval theorem of Bank, Bary-Soroker and Rosenzweig (2015); the earlier version (v5) of this preprint presented it as new, and this revision corrects that. We give a self-contained proof that the geometric monodromy group of the family is the full symmetric group  $S_{2d}$  whenever  $p = \text{char } \mathbb{F}_q > 2d$ , *without* the squarefree hypothesis on  $f$  that v5 required. We then prove an exact identity  $2d N_{\text{irr}} = \#W(\mathbb{F}_q) - E$ , where  $W$  is an explicit  $(d+1)$ -dimensional twisted form of the root variety of the family, of degree at most  $(d-1)!$ , and  $0 \leq E < q^{d+1}/(q-1)$ . Combined with the explicit Lang–Weil estimate of Cafure and Matera, this yields a fully explicit and rigorous threshold  $q_0(d)$  beyond which every Legendre interval contains an irreducible polynomial. For  $d \leq 3$  we obtain closed formulas valid for every admissible  $q$ ; for  $d = 3$  they agree with Kuz’min’s count of irreducible polynomials with two prescribed coefficients. Exhaustive and sampled computations for  $d \leq 6$  support every statement, and the code and data are public. An appendix lists the errors in v5 and how each one is fixed here.

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# 1 Introduction

## 1.1 Background

Legendre’s conjecture asserts that for every integer  $n \geq 1$  there is a prime between  $n^2$  and  $(n + 1)^2$ . It is open; the best unconditional result on prime gaps (Baker–Harman–Pintz [1]) gives primes in  $[x, x + x^{0.525}]$ , whereas the Legendre interval has length about  $2x^{1/2}$ .

In  $\mathbb{F}_q[t]$  the absolute value is  $|g| = q^{\deg g}$ . For monic  $f$  of degree  $d$  the polynomials  $g$  with  $|g - f^2| \leq |f|$  are exactly

$$\mathcal{I}_f = \{ f^2 + s : s \in \mathbb{F}_q[t], \deg s \leq d \}, \quad |\mathcal{I}_f| = q^{d+1},$$

an interval of “length”  $|f| = |f^2|^{1/2}$  around  $f^2$ , the analogue of  $[n^2, n^2 + n] \subset [n^2, (n + 1)^2]$ . All elements of  $\mathcal{I}_f$  are monic of degree  $2d$ . The prime polynomial theorem predicts about  $|\mathcal{I}_f|/(2d)$  irreducibles.

**Definition 1.1.** Write  $\mathcal{M} = \mathbb{A}^{d+1}$  with coordinates  $\mathbf{a} = (a_0, \dots, a_d)$  and put

$$F(t, \mathbf{a}) = f(t)^2 + \sum_{i=0}^d a_i t^i, \quad N_{\text{irr}}(f) = \#\{\mathbf{a} \in \mathbb{F}_q^{d+1} : F(t, \mathbf{a}) \text{ irreducible in } \mathbb{F}_q[t]\}.$$

## 1.2 What was already known

Bank, Bary-Soroker and Rosenzweig [2] proved that for every partition  $\lambda$  of  $k$ , every monic  $f$  of degree  $k$  and every  $m$  with  $1 \leq m < k$  (with  $m \geq 2$  if  $p \mid k(k - 1)$ , and  $m \geq 3$  if  $p = 2$  and  $\deg f' \leq 1$ ), the number of  $g \in f + \mathbb{F}_q[t]_{\leq m}$  with factorization type  $\lambda$  is  $P(\lambda)q^{m+1} + O_k(q^{m+1/2})$ , where  $P(\lambda)$  is the proportion of permutations of cycle type  $\lambda$  in  $S_k$ . Taking  $k = 2d$ ,  $m = d$  and replacing  $f$  by  $f^2$  gives, for every odd  $q$ ,

$$N_{\text{irr}}(f) = \frac{q^{d+1}}{2d} + O_d(q^{d+1/2}). \quad (1)$$

This is exactly the “Main Theorem” of v5 of this preprint, which therefore is not new. The implied constant in [2] is not explicit.

## 1.3 Results of this paper

Throughout,  $q$  is a power of a prime  $p$ , and  $f \in \mathbb{F}_q[t]$  is monic of degree  $d \geq 1$ . We write  $G_{\text{geom}}$  for the Galois group of  $F(t, \mathbf{a})$  over  $\overline{\mathbb{F}}_q(\mathbf{a})$ , i.e. the geometric monodromy group of the finite cover  $\{F = 0\} \rightarrow \mathcal{M}$  (Section 3).

**Theorem A** (Section 3). *If  $p > 2d$ , then  $G_{\text{geom}} = S_{2d}$ . No assumption on  $f$  beyond monicity is needed.*

**Theorem B** (Section 4). *There is an affine variety  $W$  over  $\mathbb{F}_q$ , explicitly defined by  $d - 1$  equations of degrees  $1, 2, \dots, d - 1$  in  $\mathbb{A}^{2d}$ , such that for all  $q$*

$$2d \cdot N_{\text{irr}}(f) = \#W(\mathbb{F}_q) - E(f), \quad 0 \leq E(f) \leq \sum_{e|2d, e < 2d} q^e < \frac{q^{d+1}}{q-1}.$$

*Its  $\mathbb{F}_q$ -points are the  $\beta \in \mathbb{F}_{q^{2d}}$  whose characteristic polynomial lies in  $\mathcal{I}_f$ .  $W$  has pure dimension  $d + 1$  and degree at most  $(d - 1)!$ , and it is absolutely irreducible if and only if  $G_{\text{geom}} = S_{2d}$ .*

**Theorem C** (Section 5). *Assume  $G_{\text{geom}} = S_{2d}$  (for instance  $p > 2d$ ). Let  $\delta = (d - 1)!$ . If  $q > 2(d + 2)\delta^2$ , then*

$$|2d N_{\text{irr}}(f) - q^{d+1}| < (\delta - 1)(\delta - 2) q^{d+1/2} + (5\delta^{13/3} + 2)q^d.$$

*Consequently  $N_{\text{irr}}(f) \geq 1$  for all  $q > q_0(d) := \max\{2(d + 2)\delta^2, ((\delta - 1)(\delta - 2) + \sqrt{5\delta^{13/3} + 2})^2\}$ .*

**Theorem D** (Section 6). *(i) If  $d = 1$  then  $N_{\text{irr}} = (q^2 - q)/2$ . (ii) If  $d = 2$  and  $q$  is odd then  $N_{\text{irr}} = (q^3 - q)/4$ . (iii) If  $d = 3$  and  $p \geq 5$ , write  $f = t^3 + c_2 t^2 + c_1 t + c_0$  and  $\Delta = c_2^2 - 3c_1$ ; let  $\eta$  be the quadratic character of  $\mathbb{F}_q$ . Then*

$$6 N_{\text{irr}} = \begin{cases} q^4 - \eta(2\Delta) q^2 - q + \eta(-3) + \eta(2\Delta) - 1, & \Delta \neq 0, \\ q^4 - q - (q - 1) \eta(-3), & \Delta = 0. \end{cases}$$

*In particular every Legendre interval with  $d \leq 3$  contains an irreducible polynomial, for every  $q$  in the stated range.*

Theorem C gives, for example,  $q_0(4) \approx 1.65 \times 10^4$  and  $q_0(5) \approx 7.3 \times 10^6$  (Table 1). For  $d \leq 3$  no threshold is needed. Theorem D(i)–(ii) are classical consequences of the counts of irreducible polynomials with prescribed trace [5], and (iii) is a special case of Kuz'min's formula for two prescribed coefficients (see [13, 15]). We include short proofs because they come out of Theorem B directly and they make good test cases for the computations.

The genuinely hard analogue of Legendre's conjecture fixes  $q$  and lets  $d \rightarrow \infty$ . Then one must prescribe the top  $d - 1$  coefficients of a polynomial of degree  $2d$ , about half of them. Existing results on irreducible polynomials with prescribed coefficients [17, 10] handle far fewer coefficients, so this regime remains open (Section 8).

## 1.4 About this revision

Version v5 [6] contained several gaps and incorrect citations, listed in Appendix A. The most serious were: the main theorem was not new; the primitivity argument used a criterion that does not apply as stated; the effective Chebotarev bound was formulated with the wrong sheaf; and the explicit threshold  $(8d)^{2d+6}$  relied on a Betti-number estimate that is not in the cited source. The present version replaces the cohomological error analysis by the elementary twisted-variety identity (Theorem B) and an explicit Lang–Weil bound, so that every constant is proved.

## 2 Preliminaries

For  $\beta \in \mathbb{F}_{q^n}$  let  $\chi_\beta(t) = \prod_{k=0}^{n-1} (t - \beta^{q^k}) \in \mathbb{F}_q[t]$  be its characteristic polynomial over  $\mathbb{F}_q$ . If  $\beta$  has degree  $e \mid n$  over  $\mathbb{F}_q$  and minimal polynomial  $\mu_\beta$ , then  $\chi_\beta = \mu_\beta^{n/e}$ .

**Lemma 2.1** (Membership in  $\mathcal{I}_f$ ). *A monic  $P \in \mathbb{F}_q[t]$  of degree  $2d$  lies in  $\mathcal{I}_f$  if and only if its coefficients of  $t^{2d-1}, \dots, t^{d+1}$  agree with those of  $f^2$ . If  $p > d - 1$ , this holds if and only if the power sums  $p_j(P) = \sum_{P(\rho)=0} \rho^j$  satisfy  $p_j(P) = 2p_j(f)$  for  $1 \leq j \leq d - 1$ .*

*Proof.* The first statement is the definition. For the second, Newton's identities express the power sums  $p_1, \dots, p_{d-1}$  through the top  $d - 1$  coefficients. The converse direction needs division by  $1, \dots, d - 1$ , which is allowed when  $p > d - 1$ . Finally  $p_j(f^2) = 2p_j(f)$ .  $\square$

**Lemma 2.2** (Inertia at a simple node). *Let  $R$  be a complete discrete valuation ring with algebraically closed residue field  $k$  of characteristic  $\neq 2$ , fraction field  $K$  and valuation  $v$ . Let  $P \in R[t]$  be monic of degree  $n$  whose reduction  $\bar{P}$  has exactly one double root  $\beta$  and  $n - 2$  simple roots, and suppose  $v(\text{disc}P) = 1$ . Then the splitting field of  $P$  over  $K$  is a quadratic extension whose Galois group acts on the roots of  $P$  as a transposition.*

*Proof.* Write  $\bar{P} = (t - \beta)^2 \bar{L}$  with  $\bar{L}(\beta) \neq 0$  and  $\bar{L}$  squarefree. By Hensel's lemma  $P = QL$  with  $Q, L \in R[t]$  monic lifting  $(t - \beta)^2$  and  $\bar{L}$ , and  $L$  splits into linear factors over  $R$ . Since  $\text{disc}P = \text{disc}Q \cdot \text{disc}L \cdot \text{Res}(Q, L)^2$  and both  $\text{disc}L$  and  $\text{Res}(Q, L)$  are units,  $v(\text{disc}Q) = 1$ . An element of odd valuation is not a square in  $K$ , so  $Q$  is irreducible over  $K$  and its splitting field is a quadratic extension. Its nontrivial automorphism swaps the two roots of  $Q$  and fixes the roots of  $L$ .  $\square$

**Lemma 2.3** (Primitivity and decomposition; cf. [18, 16]). *Let  $K$  be a field,  $G \in K[t]$  of degree  $n \geq 2$  with  $G' \neq 0$ , and  $x$  transcendental over  $K$ . If  $\text{Gal}(G(t) - x / K(x))$ , acting on the roots of  $G(t) - x$ , is imprimitive, then  $G = g \circ h$  with  $g, h \in K[t]$  and  $\deg g, \deg h \geq 2$ .*

*Proof.* Let  $\tau$  be a root and  $\Gamma$  the Galois group. Blocks containing  $\tau$  correspond to subgroups between the stabilizer of  $\tau$  and  $\Gamma$ , hence to fields  $K(x) \subseteq L \subseteq K(\tau)$ . Imprimitivity gives a proper intermediate field  $L$ , and  $L = K(h)$  for some  $h \in K(\tau)$  by Lüroth's theorem. Since  $x = G(\tau)$  is a polynomial in  $\tau$ , the place  $\tau = \infty$  is the only place of  $K(\tau)$  above  $x = \infty$ ; it is  $K$ -rational. Its restriction to  $L$  is therefore the only place of  $L$  above  $x = \infty$ , and it is  $K$ -rational. After a Möbius transformation over  $K$  we may take it to be  $h = \infty$ . Then  $h$ , as a function of  $\tau$ , has its only pole at  $\tau = \infty$ , so  $h \in K[\tau]$ . Likewise  $x \in K(h)$  has its only pole at  $h = \infty$ , so  $x = g(h)$  with  $g \in K[X]$ . Both degrees are  $\geq 2$  because  $L$  is a proper intermediate field.  $\square$

### 3 The monodromy group

Let  $\mathcal{H} = \{F(t, \mathbf{a}) = 0\} \subset \mathcal{M} \times \mathbb{A}^1$  and  $\pi : \mathcal{H} \rightarrow \mathcal{M}$  the projection, a finite flat morphism of degree  $2d$ . Let  $\mathcal{D} \subset \mathcal{M}$  be the zero locus of  $\text{disc}_t F$  and  $\mathcal{M}^\circ = \mathcal{M} \setminus \mathcal{D}$ . Over  $\mathcal{M}^\circ$ ,  $\pi$  is finite étale. Its geometric monodromy group is the image of  $\pi_1(\mathcal{M}_{\mathbb{F}_q}^\circ)$  in  $S_{2d}$ , which equals  $G_{\text{geom}} = \text{Gal}(F / \mathbb{F}_q(\mathbf{a}))$ . Put  $K' = \mathbb{F}_q(a_1, \dots, a_d)$  and

$$G(t) = f(t)^2 + a_1 t + \dots + a_d t^d \in K'[t], \quad \text{so} \quad F = G(t) + a_0.$$

Since  $a_0$  is transcendental over  $K'$ ,  $G_{\text{geom}} = \text{Gal}(G(t) - x / K'(x))$  with  $x = -a_0$ .

**Proposition 3.1** (Transitivity).  *$F$  is irreducible over  $\mathbb{F}_q(\mathbf{a})$ , so  $G_{\text{geom}}$  is transitive.*

*Proof.*  $F$  has degree one in  $a_0$  with coefficients 1 and  $f^2 + \sum_{i \geq 1} a_i t^i$ , which are coprime. Hence  $F$  is irreducible in  $\mathbb{F}_q[\mathbf{a}, t]$ , and by Gauss's lemma it is irreducible in  $\mathbb{F}_q(\mathbf{a})[t]$  because it is monic in  $t$ .  $\square$

**Proposition 3.2** (Primitivity). *If  $p > d$ , then  $G_{\text{geom}}$  is primitive.*

*Proof.* For  $d = 1$  this is automatic since  $2d = 2$ . Let  $d \geq 2$ . Note  $G' \neq 0$  because of the term  $a_1 t$ . Suppose  $G_{\text{geom}}$  is imprimitive. By Lemma 2.3 applied over  $K'$ ,  $G = g \circ h$  with  $g, h \in K'[t]$ ,  $\deg h = m$ ,  $\deg g = n$ ,  $mn = 2d$  and  $m, n \geq 2$ . In particular  $m \leq d$ . Replacing  $h$  by  $\lambda h + \mu$  and  $g$  accordingly, we may assume  $h$  monic with  $h(0) = 0$ . Write  $h = t^m + h_{m-1}t^{m-1} + \dots + h_1 t$ .

*Step 1:  $h$  has constant coefficients.* For  $1 \leq j \leq m-1$  we have  $2d-j \geq 2d-m+1 \geq d+1$ , so the coefficient of  $t^{2d-j}$  in  $G$  is the corresponding coefficient of  $f^2$ , an element of  $\mathbb{F}_q$ . On the other side,  $g(h) = ch^n + (\text{terms of degree} \leq 2d-m)$  with  $c$  the leading coefficient of  $g$ , and  $c = 1$  because  $G$  is monic. Expanding  $h^n$  gives

$$[t^{2d-j}] h^n = n h_{m-j} + \Phi_j(h_{m-1}, \dots, h_{m-j+1}),$$

with  $\Phi_j$  a polynomial with integer coefficients. Since  $2 \leq n \leq d < p$ ,  $n$  is invertible in  $\mathbb{F}_q$ . Induction on  $j$  shows  $h_{m-1}, \dots, h_1 \in \mathbb{F}_q$ , so  $h \in \mathbb{F}_q[t]$ .

*Step 2: contradiction.* Let  $U = \text{span}_{\mathbb{F}_q} \{1, h, \dots, h^n\} \subset \mathbb{F}_q[t]_{\leq 2d}$ . Since  $G = \sum_k g_k h^k$  with  $g_k \in K'$ , we get  $G \in U \otimes_{\mathbb{F}_q} K'$ . Let  $\ell$  be any  $\mathbb{F}_q$ -linear form on  $\mathbb{F}_q[t]_{\leq 2d}$  that vanishes on  $U$ , extended  $K'$ -linearly. Then  $0 = \ell(G) = \ell(f^2) + \sum_{i=1}^d a_i \ell(t^i)$  in  $K'$ , and since  $a_1, \dots, a_d$  are algebraically independent,  $\ell(t) = 0$ . As this holds for every such  $\ell$ ,  $t \in U$ . But every nonconstant element of  $U$  has degree divisible by  $m \geq 2$ . Contradiction.  $\square$

**Proposition 3.3** (A transposition). *If  $p > 2d$ , then  $G_{\text{geom}}$  contains a transposition.*

*Proof. Step 1: a fibre with exactly one double root.* If  $d = 1$ , take  $P^* = (t - \beta)^2$  for any  $\beta \in \overline{\mathbb{F}}_q$ ; it lies in  $\mathcal{I}_f \otimes \overline{\mathbb{F}}_q$  since  $\deg((t - \beta)^2 - f^2) \leq 1$ . Let  $d \geq 2$  and fix  $\beta \in \overline{\mathbb{F}}_q$ . Divide  $f^2 = (t - \beta)^2 Q + r$  with  $\deg r \leq 1$ ;  $Q$  is monic of degree  $2d - 2$ . For  $c \in \overline{\mathbb{F}}_q$  put  $R_c = Q + c$  and  $P^* = (t - \beta)^2 R_c$ . Then  $P^* - f^2 = c(t - \beta)^2 - r$  has degree  $\leq 2 \leq d$ , so  $P^*$  is a fibre of  $F$  over some  $\mathbf{a}^* \in \mathcal{M}(\overline{\mathbb{F}}_q)$ . The derivative  $Q'$  has leading coefficient  $2d - 2 \not\equiv 0 \pmod{p}$ , so  $R_c$  has a repeated root only if  $c = -Q(\gamma)$  for a root  $\gamma$  of  $Q'$ . Excluding these finitely many values and  $c = -Q(\beta)$ , the polynomial  $R_c$  is squarefree and  $R_c(\beta) \neq 0$ . So  $P^* = (t - \beta)^2 R$  with  $R$  squarefree and  $R(\beta) \neq 0$  (with  $R = 1$  if  $d = 1$ ).

*Step 2: the discriminant vanishes to order one.* Consider the line  $\varepsilon \mapsto \mathbf{a}^* + \varepsilon e_0$ , along which  $F = P^* + \varepsilon$ . Since  $\partial_t F = P^{*'}$  does not depend on  $\varepsilon$  and has leading coefficient  $2d \neq 0$  in  $\mathbb{F}_q$ , we have

$$\text{disc}_t(P^* + \varepsilon) = u \prod_{\gamma: P^{*'}(\gamma)=0} (P^*(\gamma) + \varepsilon),$$

with  $u \in \overline{\mathbb{F}}_q^\times$  and roots  $\gamma$  counted with multiplicity. Its order of vanishing at  $\varepsilon = 0$  is the number of common roots of  $P^*$  and  $P^{*'}$ , counted with their multiplicity in  $P^{*'}$ . Now  $P^{*' } = (t - \beta)(2R + (t - \beta)R')$ . The root  $\beta$  is simple in  $P^{*'}$  because  $2R(\beta) \neq 0$ . A root  $\rho$  of  $R$  is not a root of  $P^{*'}$ , because  $P^{*' }(\rho) = (\rho - \beta)^2 R'(\rho) \neq 0$ . Hence the order is exactly 1. In particular the line is not contained in  $\mathcal{D}$ .

*Step 3: inertia.* The Galois group of  $F$  restricted to the line is a subgroup of  $G_{\text{geom}}$ . It contains the inertia group at  $\varepsilon = 0$ , which is the Galois group of  $P^* + \varepsilon$  over  $\overline{\mathbb{F}}_q((\varepsilon))$ . By Lemma 2.2 it is generated by a transposition.  $\square$

**Theorem 3.4** (Theorem A). *If  $p > 2d$ , then  $G_{\text{geom}} = S_{2d}$ , and also  $\text{Gal}(F/\mathbb{F}_q(\mathbf{a})) = S_{2d}$ .*

*Proof.*  $G_{\text{geom}}$  is primitive (Propositions 3.1 and 3.2) and contains a transposition (Proposition 3.3). A primitive permutation group containing a transposition is the full symmetric group [8, Thm. 3.3A]. The arithmetic group contains  $G_{\text{geom}}$  and lies in  $S_{2d}$ .  $\square$

*Remark 3.5.* (1) No squarefree hypothesis on  $f$  is used. In v5 it served only to locate a simple branch point near  $\mathbf{a} = 0$ , and that argument had gaps (Appendix A). (2) The hypothesis  $p > 2d$  is used to invert  $n \leq d$ , to make  $2d$  and  $2d - 2$  nonzero in  $\mathbb{F}_q$ , and to have  $p \neq 2$ . The result  $G_{\text{geom}} = S_{2d}$  also follows from [2, Prop. 3.6] for every odd  $p$ .

## 4 The twisted root variety

### 4.1 The root variety

Let  $e_j(u)$  denote the  $j$ -th elementary symmetric polynomial in  $u = (u_1, \dots, u_{2d})$ , and  $\gamma_j = (-1)^j [t^{2d-j}] f^2 \in \mathbb{F}_q$ . Define

$$V = \{u \in \mathbb{A}^{2d} : e_j(u) = \gamma_j, 1 \leq j \leq d-1\}.$$

Let  $\Phi : \mathbb{A}_u^{2d} \rightarrow \mathbb{A}^{2d}$  send  $u$  to the coefficients of  $\prod_i (t - u_i)$ .  $\Phi$  is finite, flat and surjective, and  $S_{2d}$  permutes the coordinates of its fibres. Identify  $\mathcal{M}$  with the affine subspace  $\mathcal{I}_f$  of monic polynomials of degree  $2d$ . Then  $V = \Phi^{-1}(\mathcal{M})$ .

**Proposition 4.1.** (a)  $V$  is a complete intersection of pure dimension  $d+1$  and degree at most  $(d-1)!$ .  
(b)  $V$  is reduced, and  $V^\circ := V \setminus \Phi^{-1}(\mathcal{D})$  is smooth and dense in  $V$ .  
(c)  $V_{\mathbb{F}_q}$  is irreducible if and only if  $G_{\text{geom}} = S_{2d}$ .

*Proof.* (a) Every component of  $V_{\mathbb{F}_q}$  has dimension at least  $2d - (d-1) = d+1$  by Krull's theorem, and at most  $\dim \mathcal{M} = d+1$  because  $\Phi$  is finite. So  $V$  is a complete intersection and hence Cohen–Macaulay. The refined Bézout inequality [9, Example 8.4.6] gives  $\deg V \leq \prod_{j < d} j = (d-1)!$ .

(b) The Jacobian of  $\Phi$  is the Vandermonde determinant, so  $\Phi$  is étale where the  $u_i$  are distinct. Hence  $V^\circ \rightarrow \mathcal{M}^\circ$  is étale and  $V^\circ$  is smooth.  $\Phi^{-1}(\mathcal{D}) \cap V$  has dimension  $\leq d$ , so it contains no component of  $V$ . Thus  $V$  is generically reduced and Cohen–Macaulay, hence reduced.

(c) Over  $\mathcal{M}^\circ$ ,  $V^\circ$  is an  $S_{2d}$ -torsor:  $S_{2d}$  acts simply transitively on each fibre, the orderings of the  $2d$  distinct roots. Choosing one ordering of the base fibre identifies it with  $S_{2d}$ , and  $\pi_1(\mathcal{M}_{\mathbb{F}_q}^\circ)$  acts by left multiplication through its image  $G_{\text{geom}}$ . Hence  $V_{\mathbb{F}_q}^\circ$  has  $[S_{2d} : G_{\text{geom}}]$  connected components. Since  $V^\circ$  is smooth and dense,  $V_{\mathbb{F}_q}$  is irreducible if and only if this index is 1.  $\square$

### 4.2 The twist

Fix an  $\mathbb{F}_q$ -basis  $\omega_1, \dots, \omega_{2d}$  of  $\mathbb{F}_{q^{2d}}$  and let  $M = (\omega_i^{q^{k-1}})_{k,i}$  be the Moore matrix, which is invertible. Define  $W \subset \mathbb{A}_x^{2d}$  by the equations  $e_j(Mx) = \gamma_j$ ,  $1 \leq j \leq d-1$ .

**Lemma 4.2.**  $W$  is defined over  $\mathbb{F}_q$ , and  $x \mapsto Mx$  is an isomorphism  $W_{\mathbb{F}_{q^{2d}}} \cong V_{\mathbb{F}_{q^{2d}}}$ . The map  $x \mapsto \beta = \sum_i x_i \omega_i$  is a bijection

$$W(\mathbb{F}_q) \xrightarrow{\sim} \{\beta \in \mathbb{F}_{q^{2d}} : \chi_\beta \in \mathcal{I}_f\}.$$

*Proof.* Put  $u_k = \sum_i \omega_i^{q^{k-1}} x_i$ . Raising the coefficients of  $u_k$  to the  $q$ -th power gives  $u_{k+1}$ , indices mod  $2d$ . So Frobenius on coefficients permutes  $u_1, \dots, u_{2d}$  cyclically and fixes each  $e_j(u)$ . Hence  $e_j(Mx) \in \mathbb{F}_q[x]$ . The second claim is clear because  $M$  is invertible over  $\mathbb{F}_{q^{2d}}$ . For  $x \in \mathbb{F}_q^{2d}$  we have  $u = (\beta, \beta^q, \dots, \beta^{q^{2d-1}})$ , and  $e_j(u)$  is  $(-1)^j$  times the coefficient of  $t^{2d-j}$  in  $\chi_\beta$ . Apply Lemma 2.1.  $\square$

**Theorem 4.3** (Theorem B). For every monic  $f$  of degree  $d \geq 1$  and every  $q$ ,

$$2d \cdot N_{\text{irr}}(f) = \#W(\mathbb{F}_q) - E(f), \quad E(f) = \#\left\{\beta \in \bigcup_{e|2d, e < 2d} \mathbb{F}_{q^e} : \chi_\beta \in \mathcal{I}_f\right\}.$$

Moreover  $0 \leq E(f) \leq \sum_{e|2d, e < 2d} q^e < q^{d+1}/(q-1)$ .  $W$  is an affine  $\mathbb{F}_q$ -variety of dimension  $d+1$  and degree at most  $(d-1)!$ , and it is absolutely irreducible if and only if  $G_{\text{geom}} = S_{2d}$ .

*Proof.* If  $\beta$  generates  $\mathbb{F}_{q^{2d}}$ , then  $\chi_\beta = \mu_\beta$  is irreducible. Conversely, an irreducible  $P \in \mathcal{I}_f$  has exactly  $2d$  roots in  $\mathbb{F}_{q^{2d}}$ , and  $\chi_\beta = P$  for each of them. Together with Lemma 4.2, this gives the identity. The bound on  $E$  comes from  $\#\mathbb{F}_{q^e} = q^e$  and the fact that proper divisors of  $2d$  are at most  $d$ . The remaining statements follow from Proposition 4.1 and Lemma 4.2, because a linear change of coordinates preserves dimension, degree and absolute irreducibility.  $\square$

*Remark 4.4.* Theorem B is a twisted form of the Chebotarev argument. It counts the Frobenius class of a  $2d$ -cycle by counting rational points on one twist of the  $S_{2d}$ -torsor  $V^\circ$ . No étale cohomology of local systems is needed, and the error term reduces to a point count on an explicit, absolutely irreducible complete intersection.

## 5 An explicit bound

**Theorem 5.1** (Cafure–Matera [4, Thm. 7.1]). *Let  $V \subset \mathbb{A}^n$  be an absolutely irreducible  $\mathbb{F}_q$ -variety of dimension  $r > 0$  and degree  $\delta$ . If  $q > 2(r+1)\delta^2$ , then  $|\#V(\mathbb{F}_q) - q^r| < (\delta-1)(\delta-2)q^{r-1/2} + 5\delta^{13/3}q^{r-1}$ .*

**Theorem 5.2** (Theorem C). *Assume  $G_{\text{geom}} = S_{2d}$ , for example  $p > 2d$ . Let  $\delta = (d-1)!$ . If  $q > 2(d+2)\delta^2$ , then*

$$|2d N_{\text{irr}}(f) - q^{d+1}| < (\delta-1)(\delta-2)q^{d+1/2} + (5\delta^{13/3} + 2)q^d.$$

*Proof.* By Theorem B,  $W$  is absolutely irreducible of dimension  $d+1$  and degree  $\delta_W \leq \delta$ . The bound in Theorem 5.1 and its hypothesis are monotone in the degree, so we may use  $\delta$  in place of  $\delta_W$ . Then  $|2d N_{\text{irr}} - q^{d+1}| \leq |\#W(\mathbb{F}_q) - q^{d+1}| + E(f)$ , and  $E(f) < q^{d+1}/(q-1) \leq 2q^d$ .  $\square$

**Corollary 5.3.** *Under the hypotheses of Theorem 5.2,  $N_{\text{irr}}(f) \geq 1$  for every*

$$q > q_0(d) = \max\left\{2(d+2)\delta^2, ((\delta-1)(\delta-2) + \sqrt{5\delta^{13/3} + 2})^2\right\}.$$

*Proof.* Put  $A = (\delta-1)(\delta-2)$ ,  $B = 5\delta^{13/3} + 2$  and  $s = \sqrt{q}$ . We need  $s^2 - As - B > 0$ . The left side increases for  $s > A/2$ , and at  $s = A + \sqrt{B}$  it equals  $A\sqrt{B} \geq 0$ .  $\square$

A second explicit route uses étale cohomology directly, with Katz’s explicit bound on sums of Betti numbers.

**Proposition 5.4.** *Assume  $G_{\text{geom}} = S_{2d}$  and put  $\sigma_d = 3(d+1)^{3d-1}$ . Then for every  $q$ ,*

$$|2d N_{\text{irr}}(f) - q^{d+1}| \leq (\sigma_d - 1)q^{d+1/2} + 2q^d,$$

*and hence  $N_{\text{irr}}(f) \geq 1$  for all  $q > \sigma_d^2 = 9(d+1)^{6d-2}$ .*

*Proof.*  $W \subset \mathbb{A}^{2d}$  is defined by  $d-1$  equations of degree  $\leq d-1$ . By [12, Theorem 12 and the remark following it], applied with no characters, the sum of the compactly supported Betti numbers of  $W_{\overline{\mathbb{F}}_q}$  is at most  $3(2 + (d-1))^{2d+(d-1)} = \sigma_d$ . By the Grothendieck–Lefschetz trace formula,  $\#W(\mathbb{F}_q) = \sum_i (-1)^i \text{Tr}(\text{Frob} \mid H_c^i(W_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell))$ . Since  $W$  is absolutely irreducible of dimension  $d+1$ ,  $H_c^{2d+2}$  is one-dimensional with eigenvalue  $q^{d+1}$ . Every other  $H_c^i$  is mixed of weights  $\leq i \leq 2d+1$  by Deligne [7, Thm. 3.3.1], so its eigenvalues have absolute value  $\leq q^{d+1/2}$ . This gives  $|\#W(\mathbb{F}_q) - q^{d+1}| \leq (\sigma_d - 1)q^{d+1/2}$ , and we finish with  $E(f) \leq 2q^d$ . For  $s = \sqrt{q} > \sigma_d$  we have  $s^2 - (\sigma_d - 1)s - 2 > \sigma_d - 2 > 0$ .  $\square$

*Remark 5.5.* Proposition 5.4 needs no lower bound on  $q$  for the estimate itself, but its threshold is larger than that of Corollary 5.3 for every  $d$  (Table 1). Both rigorous thresholds have the shape  $d^{O(d)}$ , like the unproved v5 value  $(8d)^{2d+6}$ . Corollary 5.3 is smaller than the v5 value for every  $d \leq 52$ ; since  $\log q_0(d) \sim \frac{13}{3} \log(d-1)!$  while  $\log(8d)^{2d+6} \sim 2d \log d$ , the order reverses for large  $d$ . A sharper Betti-number bound for  $W$  would improve both.

| $d$ | v5 claim (unproved) | Katz + Deligne | this paper (Cor. 5.3) |
|-----|---------------------|----------------|-----------------------|
| 1   | 7.2                 | 2.2            | 0.8*                  |
| 2   | 12.0                | 5.7            | 0.9*                  |
| 3   | 16.6                | 10.6           | 2.0*                  |
| 4   | 21.1                | 16.3           | 4.2                   |
| 5   | 25.6                | 22.7           | 6.9                   |
| 6   | 30.3                | 29.7           | 9.9                   |
| 8   | 39.7                | 44.8           | 16.8                  |
| 10  | 49.5                | 61.4           | 24.8                  |
| 15  | 74.9                | 106.9          | 48.1                  |
| 20  | 101.4               | 157.0          | 74.7                  |
| 30  | 157.1               | 266.4          | 134.8                 |
| 40  | 215.4               | 384.8          | 201.4                 |
| 50  | 275.8               | 509.8          | 272.8                 |

\* not needed: Theorem D gives  $N_{\text{irr}} \geq 1$  for all admissible  $q$ .

Table 1: Thresholds for  $N_{\text{irr}} \geq 1$ , as  $\log_{10} q_0(d)$ . The column “v5” is the value  $(8d)^{2d+6}$  claimed in v5; it was not proved. The column “this paper” is Corollary 5.3. For  $d \leq 3$  Theorem D removes the threshold. Source: `data/thresholds.csv`.

## 6 Exact formulas for $d \leq 3$

**Theorem 6.1** (Theorem D (i), (ii)). *If  $d = 1$  then  $N_{\text{irr}} = (q^2 - q)/2$ . If  $d = 2$  and  $q$  is odd then  $N_{\text{irr}} = (q^3 - q)/4$ , for every  $f$ .*

*Proof.* For  $d = 1$ ,  $W = \mathbb{A}^2$  and  $E = \#\mathbb{F}_q = q$ . For  $d = 2$ ,  $W$  is the affine hyperplane  $\text{Tr}_{\mathbb{F}_{q^4}/\mathbb{F}_q}(\beta) = \gamma_1$ , which is a nonzero linear condition over  $\mathbb{F}_q$ , so  $\#W(\mathbb{F}_q) = q^3$ . If  $\beta \in \mathbb{F}_{q^2}$ , then  $\chi_\beta = (\chi_\beta^{(2)})^2$  and the condition becomes  $2\text{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(\beta) = \gamma_1$ . This has exactly  $q$  solutions, since  $q$  is odd and the trace is surjective. Hence  $E = q$ .  $\square$

**Theorem 6.2** (Theorem D (iii)). *Let  $d = 3$ ,  $p \geq 5$ ,  $f = t^3 + c_2t^2 + c_1t + c_0$  and  $\Delta = c_2^2 - 3c_1$ . Then  $6N_{\text{irr}}$  is given by the formula in Theorem D. In particular  $|6N_{\text{irr}} - q^4| \leq q^2 + q + 2$  and  $N_{\text{irr}} \geq 1$ .*

*Proof.* We use Lemma 2.1 with  $s_1 = p_1(f) = -c_2$  and  $s_2 = p_2(f) = c_2^2 - 2c_1$ . For a finite extension  $L/\mathbb{F}_q$  of degree  $n$  with  $p \nmid n$ , the trace form  $B_L(x, y) = \text{Tr}_{L/\mathbb{F}_q}(xy)$  is nondegenerate. Its Gram determinant is  $\det(M)^2$  for the Moore matrix  $M$ . Frobenius permutes the rows of  $M$  by an  $n$ -cycle, so  $\det M \in \mathbb{F}_q$  if and only if  $n$  is odd. Hence  $\eta(\det B_L) = (-1)^{n-1}$ . The restriction of  $B_L$  to  $H_L = \ker \text{Tr}_{L/\mathbb{F}_q}$  is nondegenerate with  $\det(B_L|_{H_L}) \equiv n \det B_L$  modulo squares, because  $L = \mathbb{F}_q \cdot 1 \perp H_L$  and  $B_L(1, 1) = n$ . We use the standard counts for a nondegenerate quadratic form  $Q$  in  $k$  variables over  $\mathbb{F}_q$  [14, Thms. 6.26, 6.27]:  $\#\{Q = b\} = q^{k-1} + q^{(k-1)/2}\eta((-1)^{(k-1)/2}b \det Q)$  if  $k$  is odd, and  $\#\{Q = b\} = q^{k-1} + v(b)q^{(k-2)/2}\eta((-1)^{k/2} \det Q)$  if  $k$  is even, where  $v(0) = q - 1$  and  $v(b) = -1$  for  $b \neq 0$ .

*Points of  $W$ .* Here  $L = \mathbb{F}_{q^6}$  and the conditions are  $\text{Tr}\beta = 2s_1$  and  $\text{Tr}\beta^2 = 2s_2$ . Write  $\beta = s_1/3 + \gamma$  with  $\gamma \in H_L$ . The second condition becomes  $\text{Tr}(\gamma^2) = 2s_2 - \frac{2}{3}s_1^2 = \frac{4}{3}\Delta$ . With  $k = 5$  and  $\eta(\det) = -\eta(6)$  we get  $\#W(\mathbb{F}_q) = q^4 - \eta(8\Delta)q^2 = q^4 - \eta(2\Delta)q^2$ .

*The correction  $E$ .* For  $\beta \in \mathbb{F}_{q^3}$  the conditions read  $\text{Tr}_3\beta = s_1$  and  $\text{Tr}_3\beta^2 = s_2$ . As above this is  $\text{Tr}_3\gamma^2 = \frac{2}{3}\Delta$  on a 2-dimensional space with  $\eta(\det) = \eta(3)$ . This gives  $E_3 = q - \eta(-3)$  if  $\Delta \neq 0$  and  $E_3 = q + (q - 1)\eta(-3)$  if  $\Delta = 0$ . For  $\beta \in \mathbb{F}_{q^2}$ , write  $\beta = s_1/3 + y\omega$  with  $\omega^q = -\omega$ , so  $\omega^2 = n_0$  is a nonsquare in  $\mathbb{F}_q$ . The conditions reduce to  $2n_0y^2 = \frac{4}{9}\Delta$ . This gives  $E_2 = 1 - \eta(2\Delta)$  if  $\Delta \neq 0$

and  $E_2 = 1$  otherwise. For  $\beta \in \mathbb{F}_q$  the conditions force  $\beta = s_1/3$  and  $\Delta = 0$ , so  $E_1 = [\Delta = 0]$ . Finally  $E = E_3 + E_2 - E_1$ , and Theorem B gives the formula.  $\square$

*Remark 6.3.* For  $d = 3$  the error is  $O(q^2) = O(q^{(d+1)/2})$ , much smaller than the  $q^{d+1/2}$  allowed by Chebotarev. The reason is that  $W$  is a quadric, whose only nontrivial cohomology sits in the middle degree. Theorem 6.2 is equivalent to Kuz'min's formula for irreducible sextics with the coefficients of  $t^5$  and  $t^4$  prescribed [13, 15]. The derivation via  $W$  is included because it is short and because the code checks it independently.

## 7 Numerical verification

All computations use prime fields  $q = p$ . They were done with the code in the accompanying repository, which factors every element of  $\mathcal{I}_f$  by distinct-degree factorization. The implementation was cross-checked against trial division.

### 7.1 Exact counts

Table 2 lists a sample of exact counts; the full table has one row for each  $(d, p, f)$  with  $d \leq 6$ . For  $d \leq 3$ , every computed value agrees with Theorem D. For  $d = 4, 5, 6$  the normalized error  $(2dN_{\text{irr}} - q^{d+1})/q^d$  stays bounded, consistent with Theorem C (Figure 1).

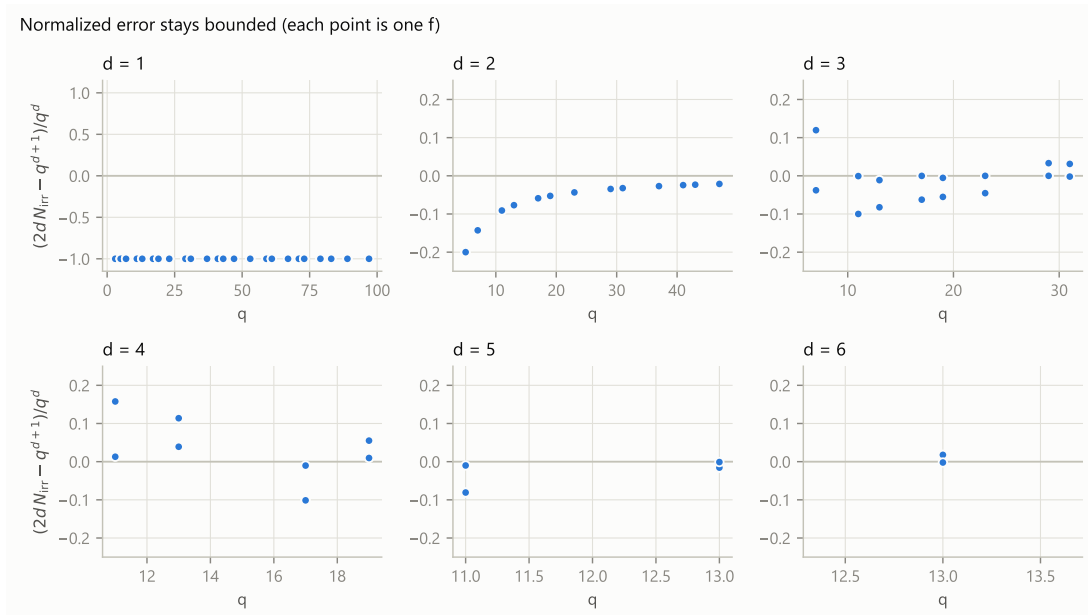


Figure 1: Normalized error  $(2dN_{\text{irr}} - q^{d+1})/q^d$  against  $q$ , for  $p > 2d$ . The values stay bounded, as Theorem C requires. For  $d = 1$  the error is exactly  $-q$ . For  $d = 2, 3$  the values tend to 0, in line with the  $O(q^{(d+1)/2})$  error of Theorem D.

### 7.2 The twisted identity

For  $d \leq 4$  and small  $p$  we enumerated all of  $\mathbb{F}_{p^{2d}}$  and counted  $\#W(\mathbb{F}_q)$  and  $E(f)$  directly. The identity of Theorem B held in every case.  $|\#W(\mathbb{F}_q) - q^{d+1}|$  was always far below the Cafure–Matera bound (data/twisted\_identity.csv).

| $d$ | $q$ | $f$                                    | $N_{\text{irr}}$ | $q^{d+1}/2d$ | $\frac{2dN - q^{d+1}}{q^d}$ | formula |
|-----|-----|----------------------------------------|------------------|--------------|-----------------------------|---------|
| 1   | 3   | $t$                                    | 3                | 4.5          | -1.000                      | ✓       |
| 1   | 43  | $t$                                    | 903              | 924.5        | -1.000                      | ✓       |
| 1   | 43  | $t + 34$                               | 903              | 924.5        | -1.000                      | ✓       |
| 1   | 97  | $t$                                    | 4656             | 4704.5       | -1.000                      | ✓       |
| 1   | 97  | $t + 93$                               | 4656             | 4704.5       | -1.000                      | ✓       |
| 2   | 5   | $t^2$                                  | 30               | 31.2         | -0.200                      | ✓       |
| 2   | 5   | $t^2 + 3t + 1$                         | 30               | 31.2         | -0.200                      | ✓       |
| 2   | 23  | $t^2$                                  | 3036             | 3041.8       | -0.043                      | ✓       |
| 2   | 23  | $t^2 + 8t + 7$                         | 3036             | 3041.8       | -0.043                      | ✓       |
| 2   | 47  | $t^2$                                  | 25944            | 25955.8      | -0.021                      | ✓       |
| 2   | 47  | $t^2 + 6t + 12$                        | 25944            | 25955.8      | -0.021                      | ✓       |
| 3   | 7   | $t^3$                                  | 398              | 400.2        | -0.038                      | ✓       |
| 3   | 7   | $t^3 + 5t^2 + 4t + 6$                  | 407              | 400.2        | +0.120                      | ✓       |
| 3   | 19  | $t^3$                                  | 21714            | 21720.2      | -0.005                      | ✓       |
| 3   | 19  | $t^3 + 11t^2 + 6t + 3$                 | 21657            | 21720.2      | -0.055                      | ✓       |
| 3   | 31  | $t^3$                                  | 153910           | 153920.2     | -0.002                      | ✓       |
| 3   | 31  | $t^3 + 7t^2 + 19t + 24$                | 154075           | 153920.2     | +0.031                      | ✓       |
| 4   | 11  | $t^4$                                  | 20420            | 20131.4      | +0.158                      |         |
| 4   | 11  | $t^4 + 5t^2 + 7t + 5$                  | 20155            | 20131.4      | +0.013                      |         |
| 4   | 17  | $t^4$                                  | 176424           | 177482.1     | -0.101                      |         |
| 4   | 17  | $t^4 + 5t^3 + 15t^2 + 2t + 4$          | 177375           | 177482.1     | -0.010                      |         |
| 4   | 19  | $t^4$                                  | 310410           | 309512.4     | +0.055                      |         |
| 4   | 19  | $t^4 + 18t^3 + 11t^2 + 6t + 12$        | 309669           | 309512.4     | +0.010                      |         |
| 5   | 11  | $t^5$                                  | 175854           | 177156.1     | -0.081                      |         |
| 5   | 11  | $t^5 + 8t^4 + 4t^3 + 8t^2 + t + 2$     | 176995           | 177156.1     | -0.010                      |         |
| 5   | 13  | $t^5$                                  | 482100           | 482680.9     | -0.016                      |         |
| 5   | 13  | $t^5 + t^4 + 7t^3 + 4t^2 + 9t + 12$    | 482642           | 482680.9     | -0.001                      |         |
| 6   | 13  | $t^6$                                  | 5236204          | 5229043.1    | +0.018                      |         |
| 6   | 13  | $t^6 + 7t^5 + 5t^4 + 9t^3 + 5t^2 + 12$ | 5228155          | 5229043.1    | -0.002                      |         |

Table 2: Exact values of  $N_{\text{irr}}(f)$  for  $f = t^d$  and a random  $f$ . “Formula” is Theorem D where it applies. Source: `data/interval_counts.csv`.

### 7.3 Factorization statistics

By Theorem A and Chebotarev, the proportion of  $\mathbf{a}$  for which  $F(t, \mathbf{a})$  has factorization type  $\lambda$  tends to  $P(\lambda) = 1/z_\lambda$ . Figure 2 compares the observed frequencies with  $P(\lambda)$  for  $(d, p) = (2, 47), (3, 31), (4, 19), (5, 13), (6, 13)$ .

### 7.4 Exhaustive search over $f$

For small  $p$  we ran over every class of  $f$ , using translation  $t \mapsto t + b$  to set  $c_{d-1} = 0$  when  $p \nmid d$ . Table 3 gives the minimum of  $N_{\text{irr}}$ . It is positive in every case, including small characteristics  $p \leq 2d$ , which are outside Theorems A and C.

## 8 Discussion

**Large  $q$  versus large  $d$ .** Theorems A–D concern fixed  $d$  and growing  $q$ . This is the regime where the Riemann hypothesis for varieties over finite fields gives square-root cancellation. The closer analogue of the classical conjecture fixes  $q$  and lets  $d \rightarrow \infty$ . That requires an irreducible

Factorization types in  $\mathcal{I}_{t^d}$  follow the cycle types of  $S_{2d}$

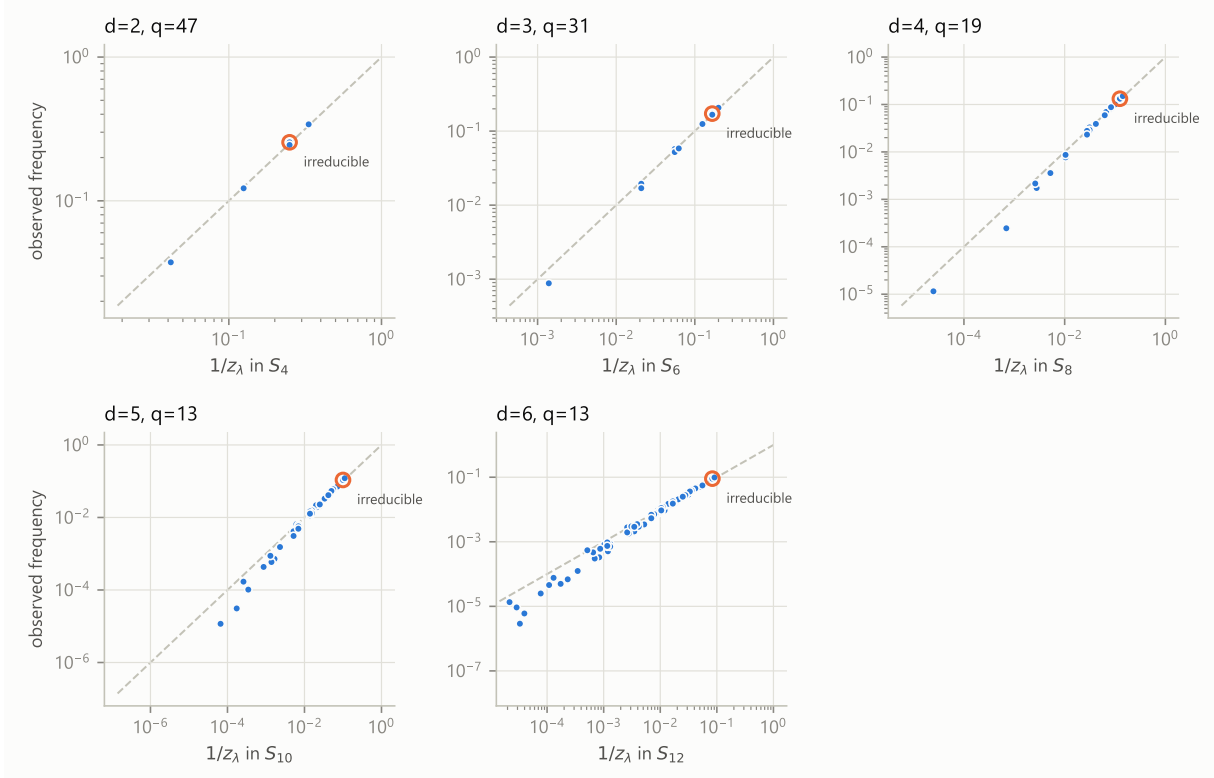


Figure 2: Observed frequency of each factorization type in  $\mathcal{I}_{t^d}$  against the  $S_{2d}$  cycle-type probability  $1/z_\lambda$ . The ringed point is the irreducible type. Points lie near the diagonal, as full  $S_{2d}$  monodromy predicts. The rarest types, with many small factors, fall below it at these small  $q$ ; this is the  $O(q^{-1/2})$  correction.

polynomial of degree  $2d$  with its top  $d - 1$  coefficients prescribed. Known results on prescribed coefficients [17, 10] allow roughly  $n/4$  prescribed coefficients in degree  $n$  for large  $q$ , and a small positive proportion for every  $q$ . Half of the coefficients is out of reach of current methods, so we make no claim there.

**Small characteristic.** For  $p \leq 2d$  the proofs of Propositions 3.2 and 3.3 do not apply as written. By [2] the asymptotic still holds for odd  $p$ . Theorem D(ii) covers  $p = 3, d = 2$ . The data of Table 3 show no failure.

### Open problems.

- (1) Prove  $N_{\text{irr}}(f) \geq 1$  for all  $q$  when  $d \geq 4$ , closing the gap between the computations (small  $q$ ) and Corollary 5.3 (large  $q$ ).
- (2) Find an explicit threshold of the form  $q > C^{d \log d}$  or better, using Betti-number bounds for  $W$ .
- (3) Shorter intervals: find the least  $e$  such that  $\{f^2 + s : \deg s \leq e\}$  contains an irreducible for all large  $q$ . By [2] the answer is 1 or 2 for large  $q$ , so the real question is uniformity in  $q$  and  $d$ .
- (4) Higher powers: the interval  $\{f^k + s : \deg s \leq (k - 1)d\}$  for  $k \geq 3$ .

| $d$ | $q$ | $p > 2d$ | classes of $f$ | $\min N_{\text{irr}}$ | $\max N_{\text{irr}}$ | $q^{d+1}/2d$ |
|-----|-----|----------|----------------|-----------------------|-----------------------|--------------|
| 2   | 3   | no       | 1              | 6                     | 6                     | 6.8          |
| 2   | 5   | yes      | 1              | 30                    | 30                    | 31.2         |
| 2   | 7   | yes      | 1              | 84                    | 84                    | 85.8         |
| 2   | 11  | yes      | 1              | 330                   | 330                   | 332.8        |
| 2   | 13  | yes      | 1              | 546                   | 546                   | 549.2        |
| 3   | 3   | no       | 9              | 11                    | 15                    | 13.5         |
| 3   | 5   | no       | 5              | 99                    | 107                   | 104.2        |
| 3   | 7   | yes      | 7              | 391                   | 407                   | 400.2        |
| 3   | 11  | yes      | 11             | 2418                  | 2458                  | 2440.2       |
| 3   | 13  | yes      | 13             | 4730                  | 4786                  | 4760.2       |
| 4   | 3   | no       | 9              | 27                    | 37                    | 30.4         |
| 4   | 5   | no       | 25             | 375                   | 402                   | 390.6        |
| 4   | 7   | no       | 49             | 2058                  | 2131                  | 2100.9       |
| 4   | 11  | yes      | 121            | 20035                 | 20420                 | 20131.4      |
| 4   | 13  | yes      | 169            | 46212                 | 46818                 | 46411.6      |
| 5   | 3   | no       | 27             | 63                    | 79                    | 72.9         |
| 5   | 5   | no       | 625            | 1450                  | 1650                  | 1562.5       |
| 5   | 7   | no       | 343            | 11654                 | 11893                 | 11764.9      |
| 6   | 3   | no       | 243            | 156                   | 204                   | 182.2        |
| 6   | 5   | no       | 625            | 6383                  | 6629                  | 6510.4       |

Table 3: Minimum and maximum of  $N_{\text{irr}}(f)$  over all  $f$ , for small  $p$ . Source: `data/exhaustive_min_counts.csv`.

## A Errata to version v5

- E1. **Novelty.** The main theorem of v5,  $N_{\text{irr}} = q^{d+1}/(2d) + O_d(q^{d+1/2})$ , is the case  $k = 2d$ ,  $m = d$  of [2, Thm. 2.3]. That theorem also holds without the squarefree hypothesis and for every odd  $p$ . v5 cited [2] only as a possible source of sharper bounds.
- E2. **Primitivity.** v5 used the equivalence “imprimitive  $\Leftrightarrow F = g \circ h$ ” for the Galois group of  $F$  over  $\overline{\mathbb{F}}_q(\mathbf{a})$ . That equivalence holds for  $G(t) - x$  with  $x$  transcendental and appearing only in the constant term, not for a general polynomial over a field. The final dimension count also treated the coefficients of  $g$ , which lie in an algebraic extension of  $\overline{\mathbb{F}}_q(\mathbf{a})$ , as free parameters in  $\mathbb{A}^n$ . Fixed in Lemma 2.3 and Proposition 3.2.
- E3. **Transposition.** v5 claimed that near  $\mathbf{a} = 0$  the discriminant has branches  $\{P(\alpha_j) = 0\}$  along which two roots coalesce. This ignores the linear term  $P'(\alpha_j)u$  and the parameter  $a_d$ . When  $P(\alpha_j) = 0$  but  $P'(\alpha_j) \neq 0$ , the double root splits. The appeal to Picard–Lefschetz was not justified in characteristic  $p$ . The remark that  $p > 2d$  is needed for  $f'(\alpha_j) \neq 0$  is also wrong, since squarefree suffices. Fixed in Proposition 3.3, which does not need  $f$  squarefree.
- E4. **Effective Chebotarev.** The bound was stated with the permutation sheaf  $\pi_* \overline{\mathbb{Q}}_\ell$ . Traces of Frobenius on that sheaf count fixed points, so they detect linear factors, not  $2d$ -cycles. Counting  $2d$ -cycles needs all irreducible characters of  $S_{2d}$  that are nonzero on the class, which are the  $2d$  hook characters. The cited “[11], Thm. 9.2.6” is not an effective Chebotarev theorem. Replaced by Theorem B.
- E5. **Betti-number bound.** v5 attributed  $B(\mathcal{F}) \leq N(2\delta + 1)^n$  to [12, Thm. 12]. That theorem bounds Betti numbers of Artin–Schreier and Kummer sheaves  $\mathcal{L}_\psi(f) \otimes \mathcal{L}_\rho(G)$ , not of push-forwards of finite covers. The general bounds for compatible systems in [12] are not explicit.

So the threshold  $(8d)^{2d+6}$  was unproved. Replaced by Corollary 5.3.

- E6. **Error term bookkeeping.** In Step 4 of v5 the unspecified  $O_{N,n}(q^{n-1})$  term of the Chebotarev bound was replaced by a Lang–Weil count of  $\mathcal{D}(\mathbb{F}_q)$ . These are different quantities, and the latter is not needed because  $N_{\text{irr}} = N_{\text{irr}}^\circ$ .
- E7. **GOS formula.** Equation (4.1) of v5 wrote  $\chi(\mathbb{P}^1, j_! \mathcal{F}) = 2r - \sum(\text{drop}_v + \text{Sw}_v)$ . That is the formula for  $j_*$ , not  $j_!$ . The section was not used in the proof and is omitted.
- E8. **Citations.** Bary-Soroker’s paper is *Proc. AMS* 137 (2009), 73–83, not vol. 140 (2012), and it works over PAC fields rather than via Chebotarev over finite fields. Several bibliography entries were never cited. The Arnold–Gusein-Zade–Varchenko reference was dated inconsistently.
- E9. **Numerical evidence.** v5 inferred that  $N_{\text{irr}} \geq 1$  “for all  $q$  and  $d$ ” from three small examples. Section 7 replaces them with systematic data. The general claim remains open for  $d \geq 4$ .
- E10. **Version-history remarks** in the body of v5 (“erroneously used in earlier versions”) have been moved to this appendix.

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